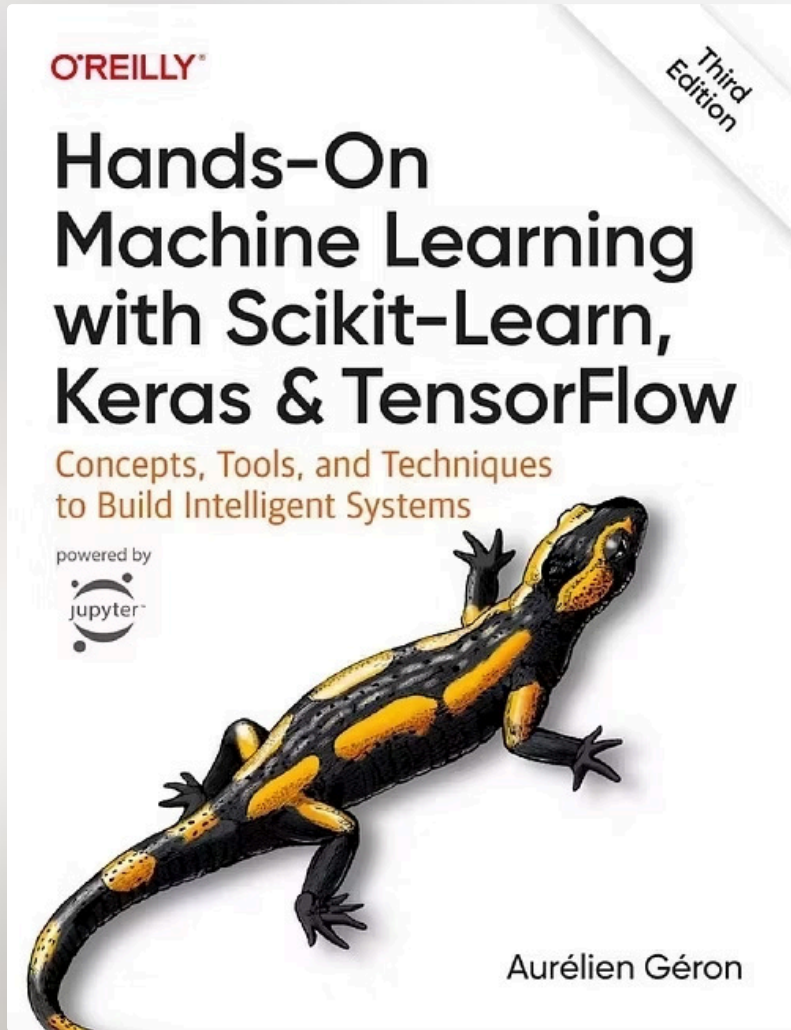




Machine Learning

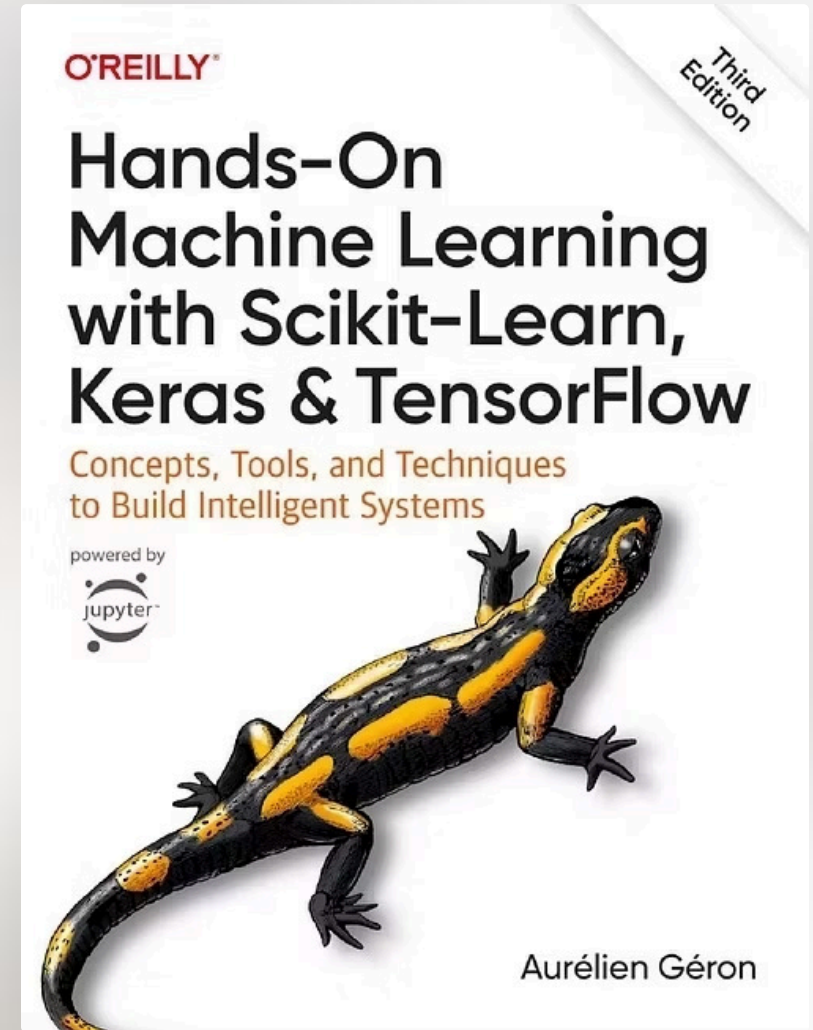
Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

RP by Rodrigo Martins Pagliares

Last update: 08/08/2025

Machine Learning

Chapter 1 – The Machine Learning Landscape



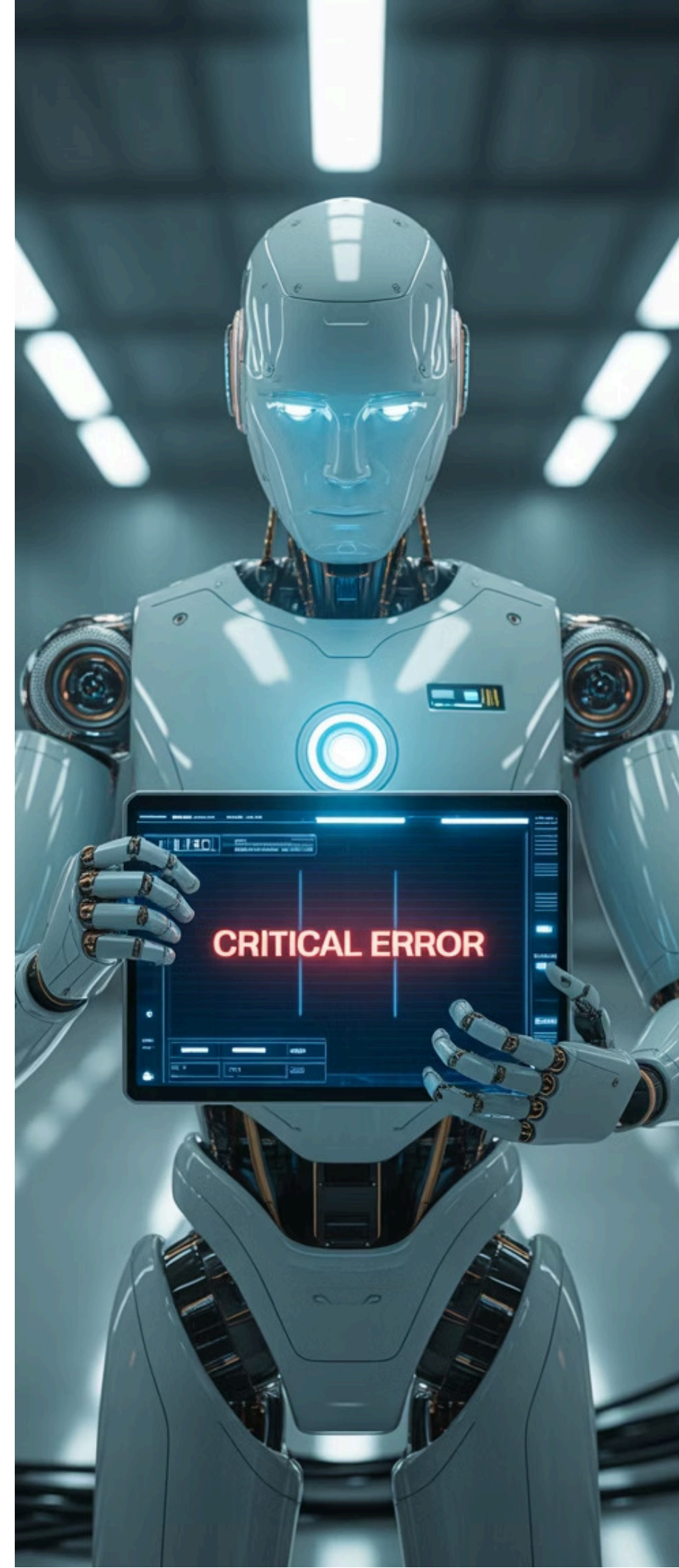


Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. **Types of Machine Learning Systems**
5. Main Challenges of Machine Learning
6. **Testing and Validating**

Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. Types of Machine Learning Systems
 - Classification criteria #1: Supervision during training
 - Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning
 - Classification criteria #2: Approach to Learning Method
 - Batch and Online (Incremental) Learning
 - Classification criteria #3: Learning Method
 - Instance-Based and Model-Based Learning
5. Main Challenges of Machine Learning
 - Insufficient Quantity of Training Data
 - Non-representative Training Data
 - Poor-Quality Data
 - Irrelevant Features
 - Overfitting the Training Data
 - Underfitting the Training Data
 - Stepping Back: Reviewing the Chapter So Far
6. Testing and Validating
 - Hyperparameter Tuning and Model Selection
 - Data Mismatch



Testing and Validating

Hyperparameter Tuning and Model Selection, **Data Mismatch**



Data Mismatch: Overview

Problem:

Data used for training may not perfectly represent the real-world data the model will encounter in production.

Example used in this lesson: Training a model with **web pictures of flowers** may not match **mobile app pictures taken by users**.

The Validation and Test Sets



1 Importance of Representativeness:

Both the **validation set** and **test set** (holdout design) must be as **representative** as possible of the **data** expected in **production**.

2 Strategy:

Shuffle the **representative data** and **split** it into the **validation set** and **test set** to ensure **no duplicates** or near-duplicates in **both sets**.

3 Challenge:

If **validation performance** is poor, it could be due to either **overfitting** the **training set** or **data mismatch** between the **web** and **mobile app images**.

Overfitting & Data mismatch Solution: Train-Dev Set

What is the Train-Dev Set?

In our example, a set of training data (from web images) is held out in a train-dev set.



Figure 1-26. When real data is scarce (right), you may use similar abundant data (left) for training and hold out some of it in a Train-dev set to evaluate overfitting; the real data is then used to evaluate data mismatch (Dev set) and to evaluate the final model's performance (Test set)

Solution: Train-Dev Set (contd.)



Train the model

On the main training set.

Evaluate on train-dev set

To check for **overfitting**.

If **performance** is **poor**, it's likely due to overfitting; you should **simplify the model** or **clean the data**.

Evaluate on dev set

(**Representative data**) to check for **data mismatch**.

If **performance** is poor here, the issue is likely due to **data mismatch**.

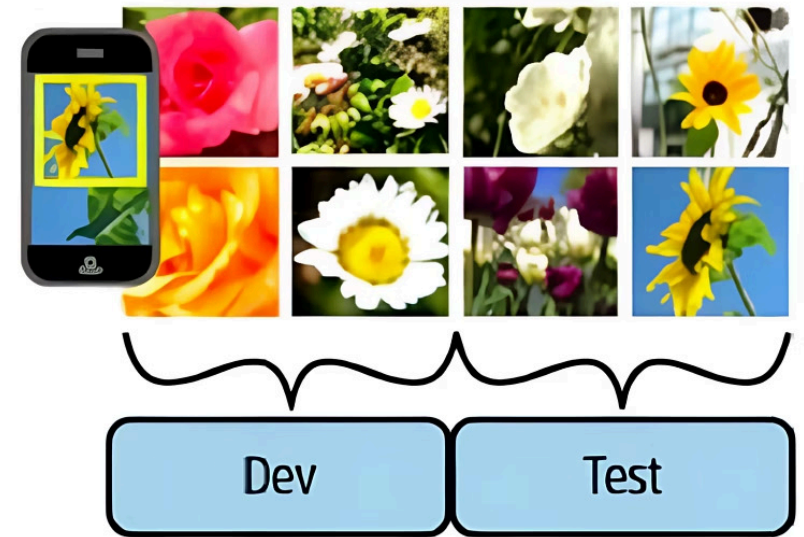
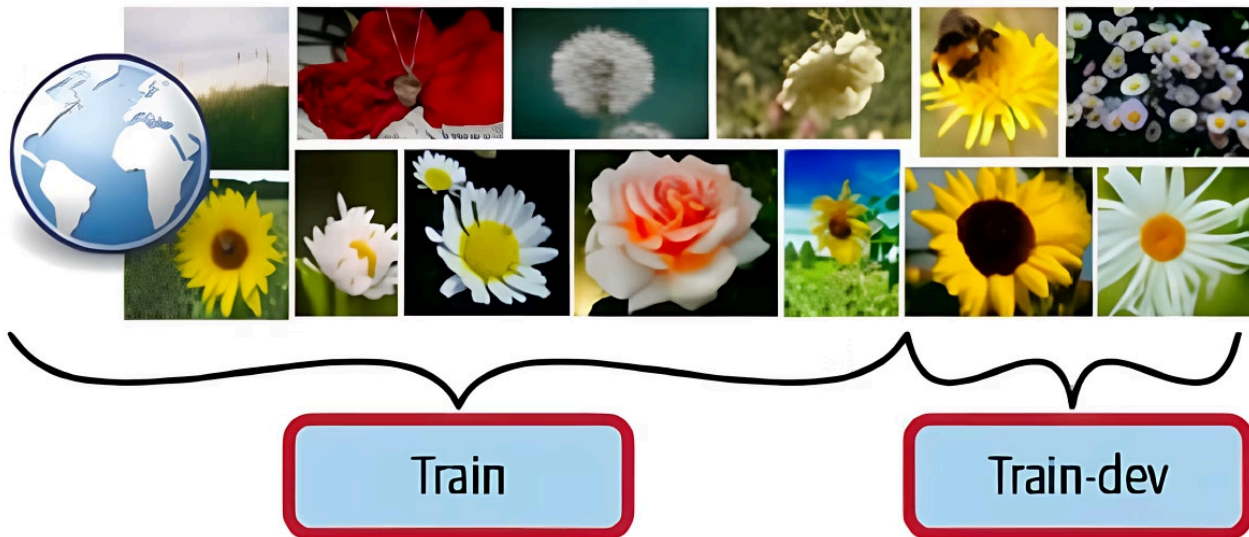
Preprocess data

Preprocess the **web images** to make them look more like **mobile app pictures**, then **retrain the model**.

Solution: Train-Dev Set (contd.)

Final Evaluation:

Once the model performs well on both the Train-dev set and the Dev set, evaluate it on the Test set to estimate how it will perform in production.



No Free Lunch Theorem

Overview:

The **No Free Lunch (NFL)** theorem states that there is **no one-size-fits-all model**.

Key Insight:

Models are **simplified representations of data**. The simplifications discard **irrelevant details** but also make **assumptions** about the **data** (e.g., a linear model assumes linear relationships).

NFL Theorem:

If **no assumptions** are made about the data, no model will **consistently outperform** others. Each model works best on different datasets.

Application:

For **simple tasks**, you might test **linear models** with various levels of **regularization**.

For **complex problems**, you might test various **neural networks**.

Key Takeaways

Data Mismatch

Ensure your **validation** and **test sets** are **representative** of real-world production data.

Use the **Train-dev set** to check for **overfitting** and **Dev set** for **data mismatch**.

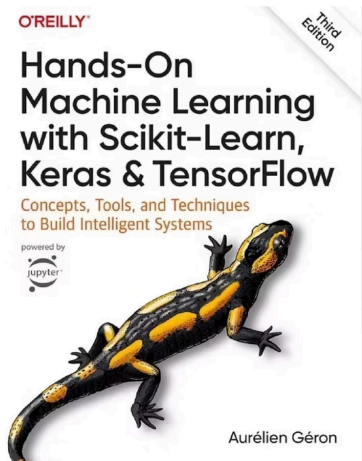


No Free Lunch Theorem

There is no universally **best model**. The best model depends on the **data** and the **problem** at hand.

Evaluate different models based on **reasonable assumptions** about the **data**.

Next: End-to-End ML Project



Chapter 2. End-to-End Machine Learning Project

In this chapter you will work through an example project end to end, pretending to be a recently hired data scientist at a real estate company. This example is fictitious; the goal is to illustrate the main steps of a machine learning project, not to learn anything about the real estate business. Here are the main steps we will walk through:

1. Look at the big picture.
2. Get the data.
3. Explore and visualize the data to gain insights.
4. Prepare the data for machine learning algorithms.
5. Select a model and train it.
6. Fine-tune your model.
7. Present your solution.
8. Launch, monitor, and maintain your system.

Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts